

Merge and Code Review (TAES I, IN0977)

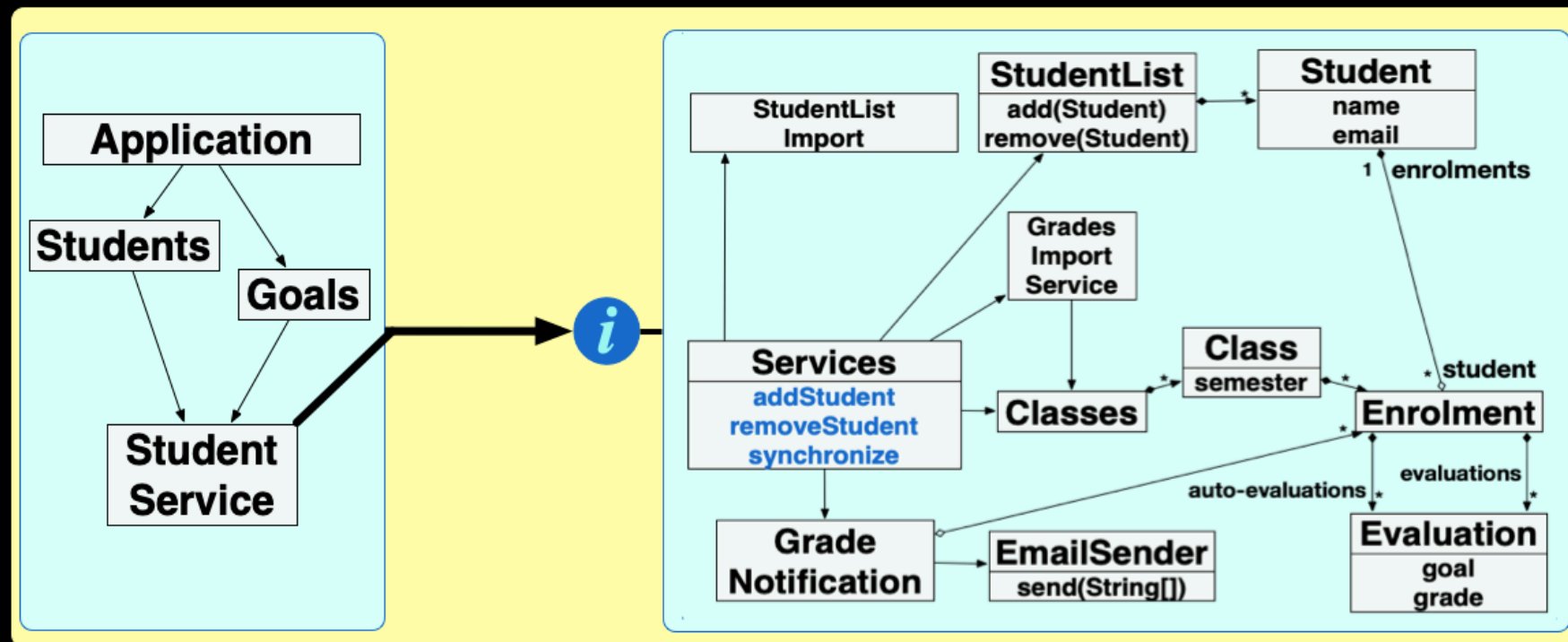
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Ferramentas de integração e revisão de código:
A disciplina apresenta e discute várias técnicas, práticas e ferramentas avançadas para integração (merge) e revisão de código, incluindo LLMs e agentes, e cobrindo artigos e estudos recentes da área, e os principais métodos de pesquisa utilizados na área. A avaliação é baseada em uma prova, roteiros de exercícios, participação em aula e um projeto. A disciplina é presencial, exceto pela parte de acompanhamento do projeto, que pode ser remota.

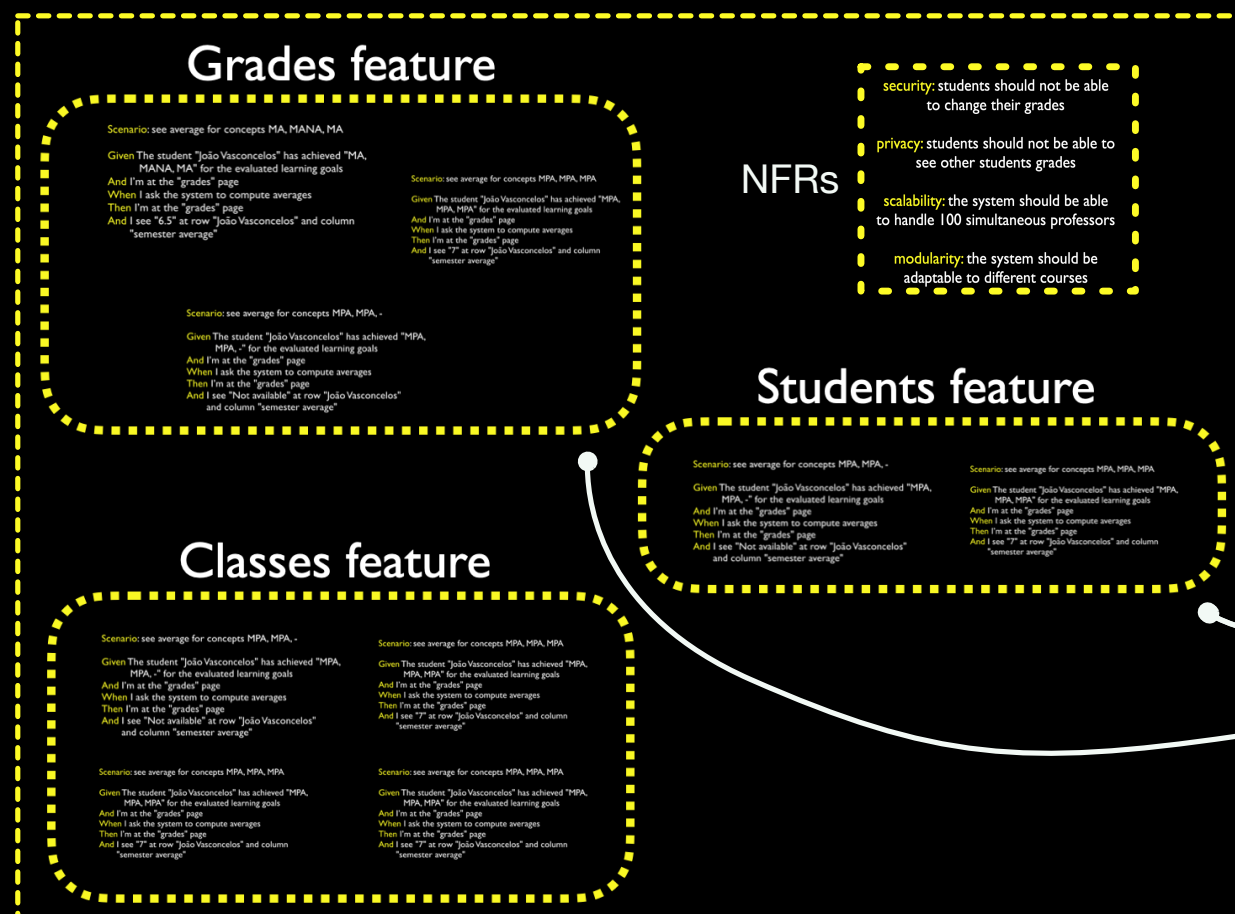
Context

Collaborative software development



Task structure is often derived from requirements structure

System requirements specification



8

To do

!

Student help analysis based on low threshold

...

#252 opened by pauloborba

bug

!

Import student list from excel file

...

#250 opened by pauloborba

enhancement

!

Final grade computation and visualization

...

#249 opened by pauloborba

enhancement

!

Class performance report

...

#248 opened by pauloborba

enhancement

!

Close subscriptions after usage

...

#251 opened by pauloborba

bug

Tasks are often crosscutting

8 To do

! Student help analysis based on low threshold

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! Import student list from excel file

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enhancement

! Final grade computation and visualization

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enhancement

! Class performance report

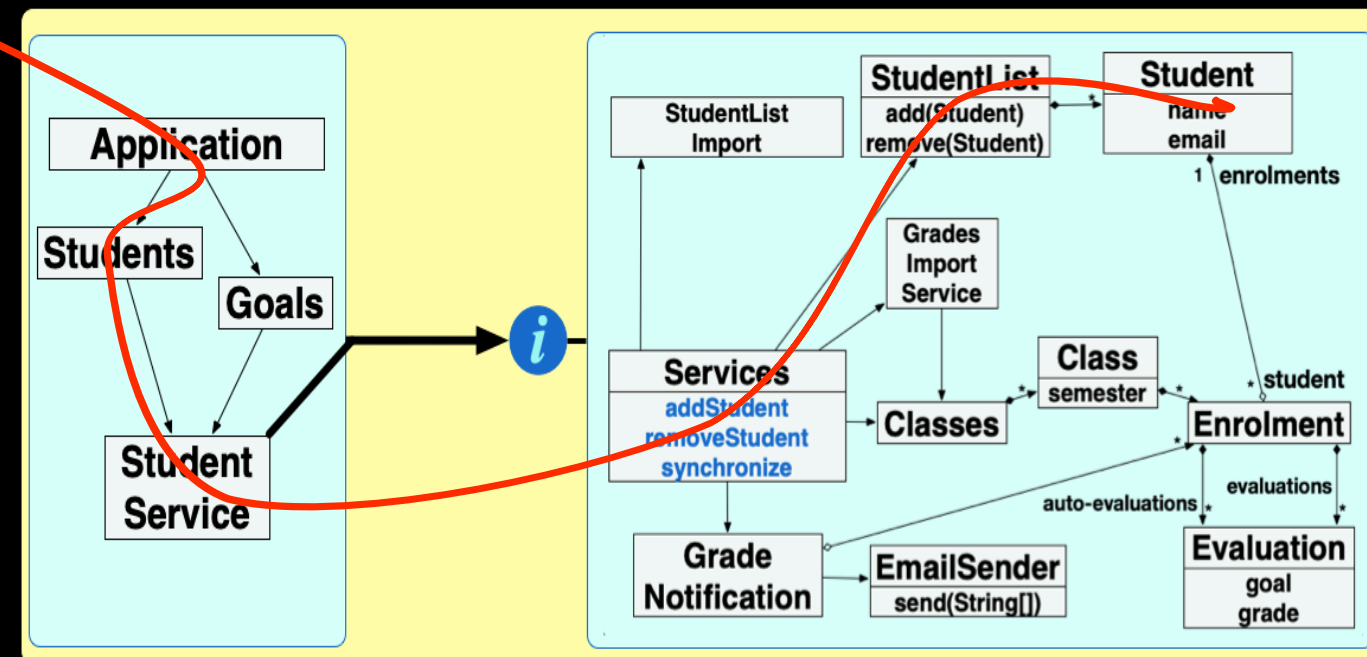
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enhancement

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bug



Tasks might involve changing classes in common

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Class performance report

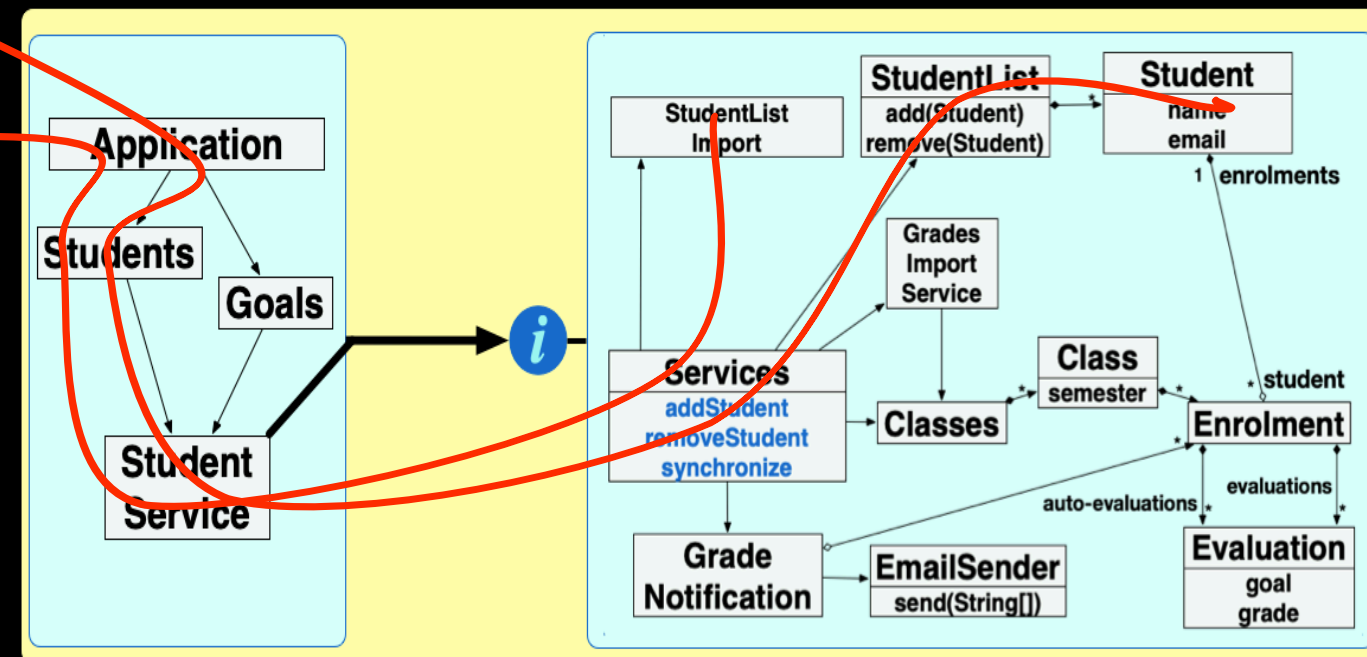
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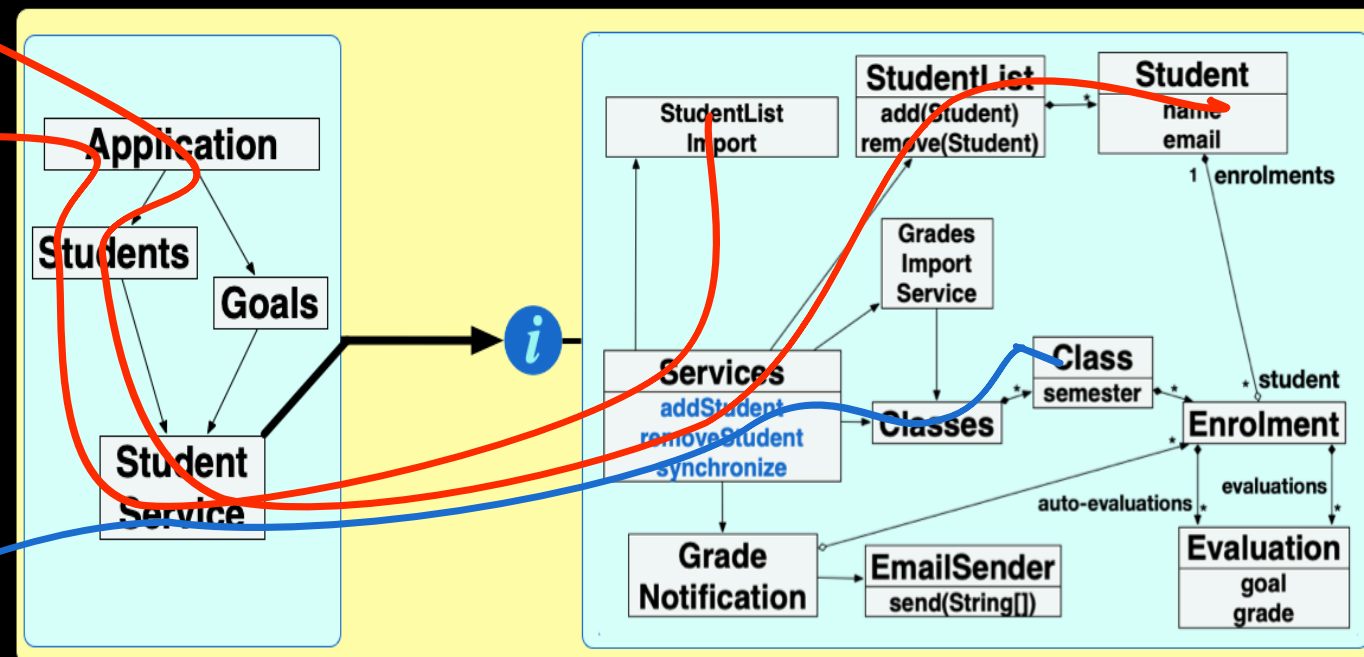
bug



Task structure often does not match code structure

8 To do + ...

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- ! Class performance report ...
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enhancement
- ! Close subscriptions after usage ...
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Modular development is not always possible, no matter the investment in modularity

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Class performance report

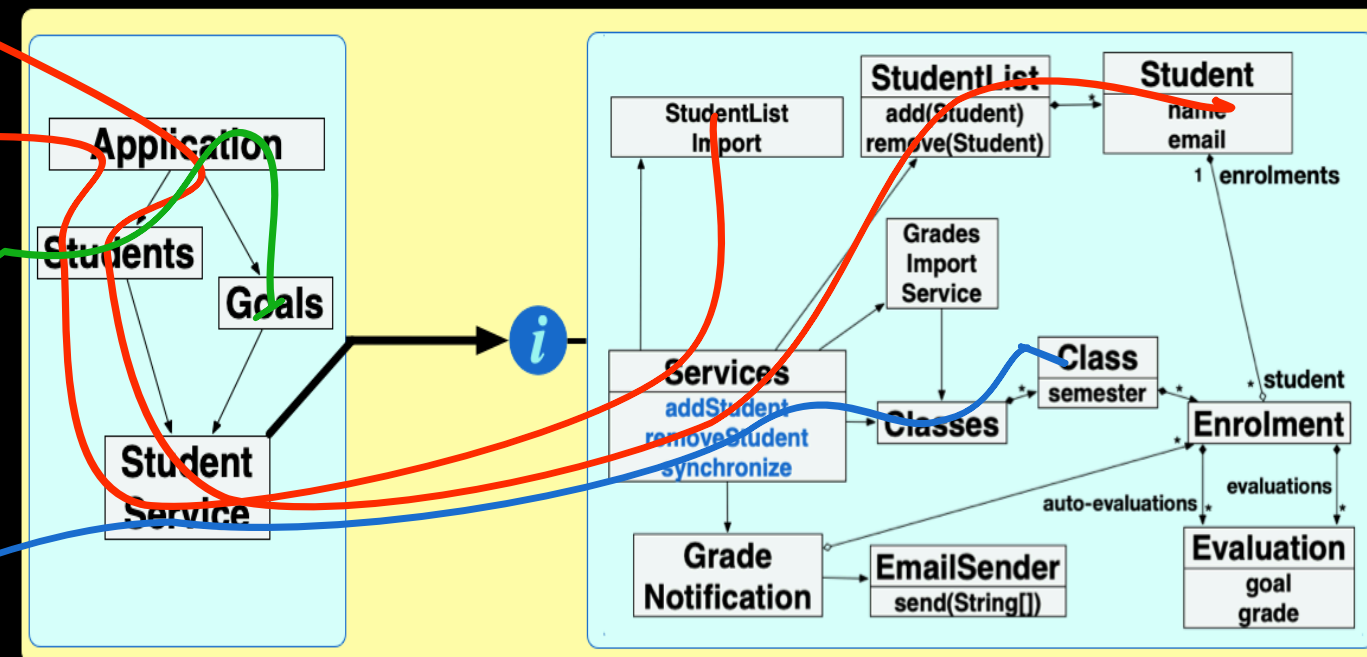
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enhancement

Close subscriptions after usage

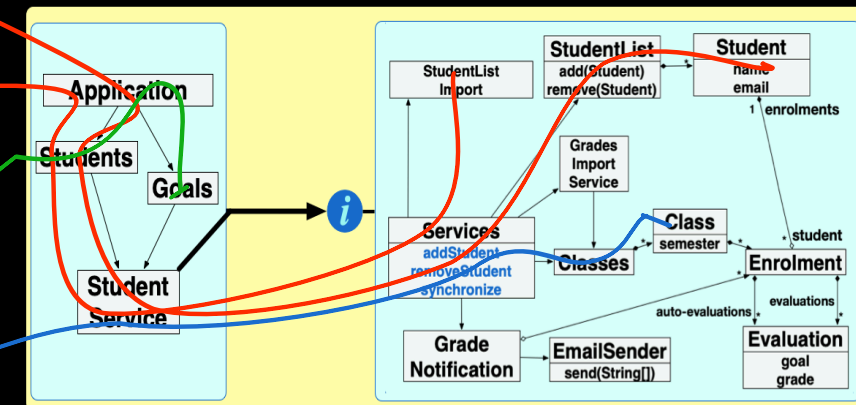
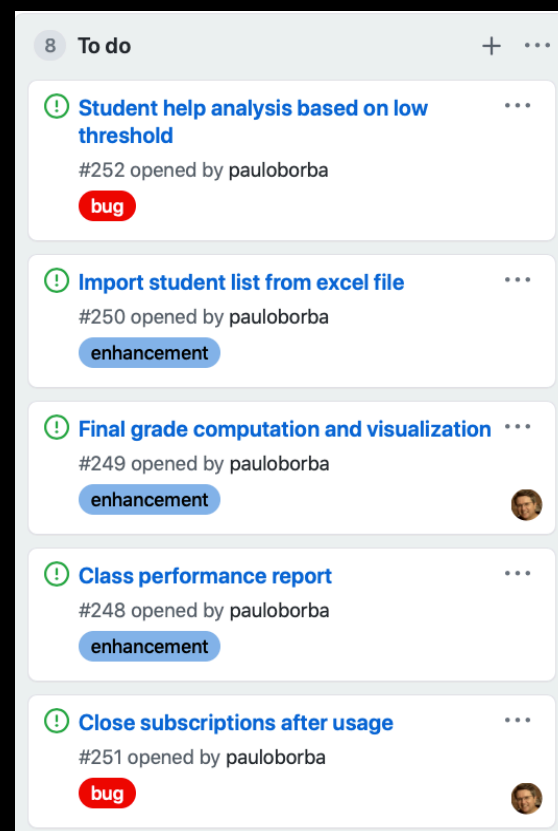
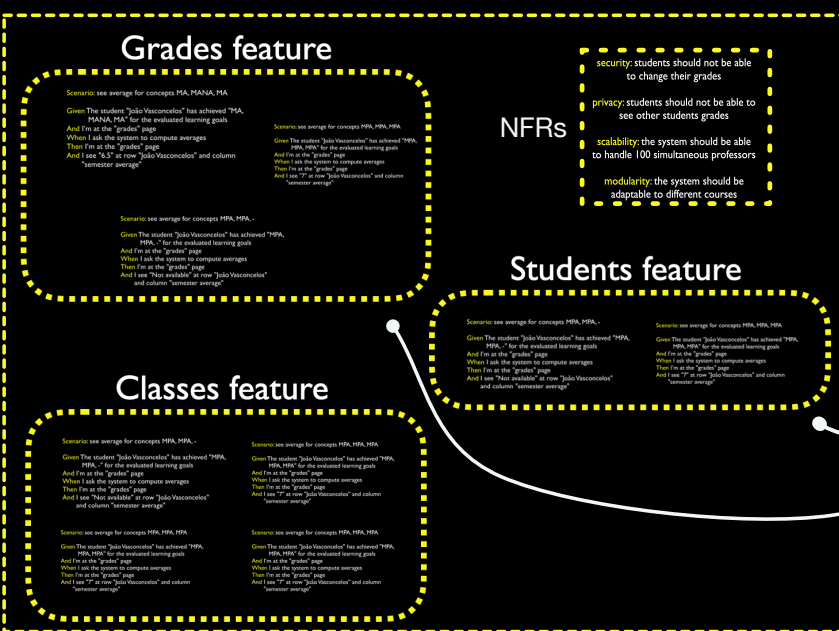
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bug



Different modular structures for different artifacts

System requirements specification



Problem

What could go wrong during code integration?

(assuming branching and merging is available)

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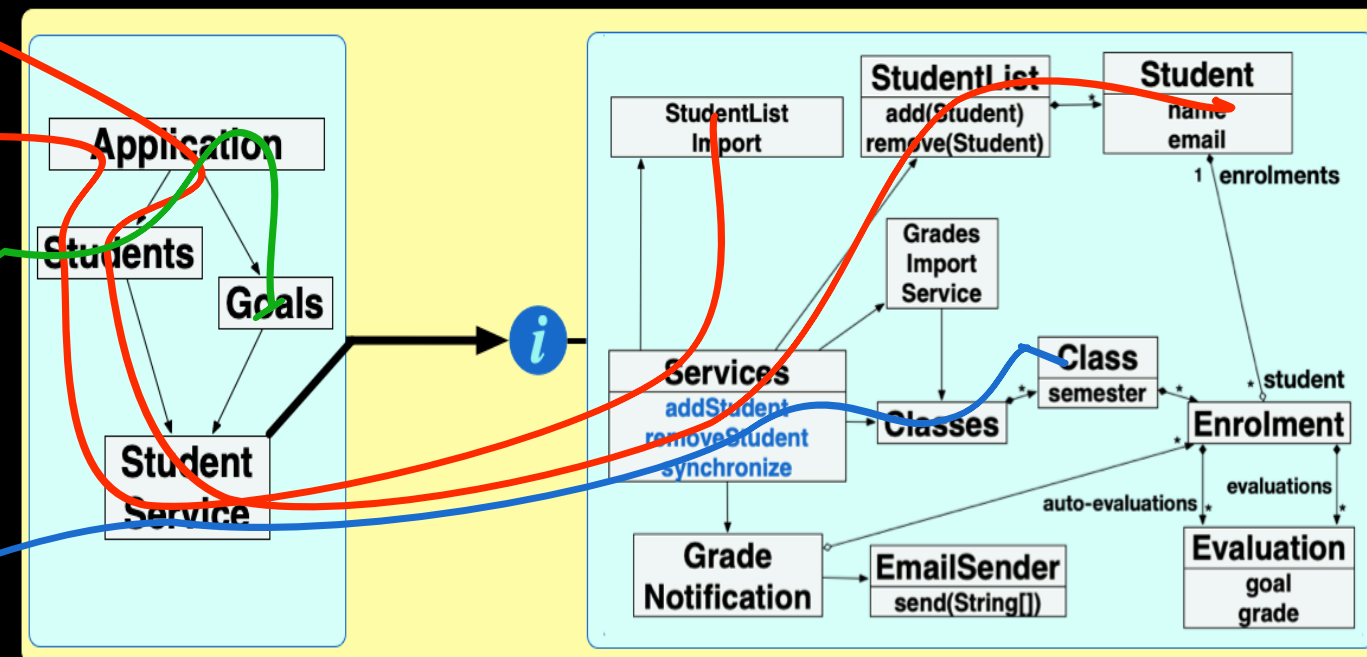
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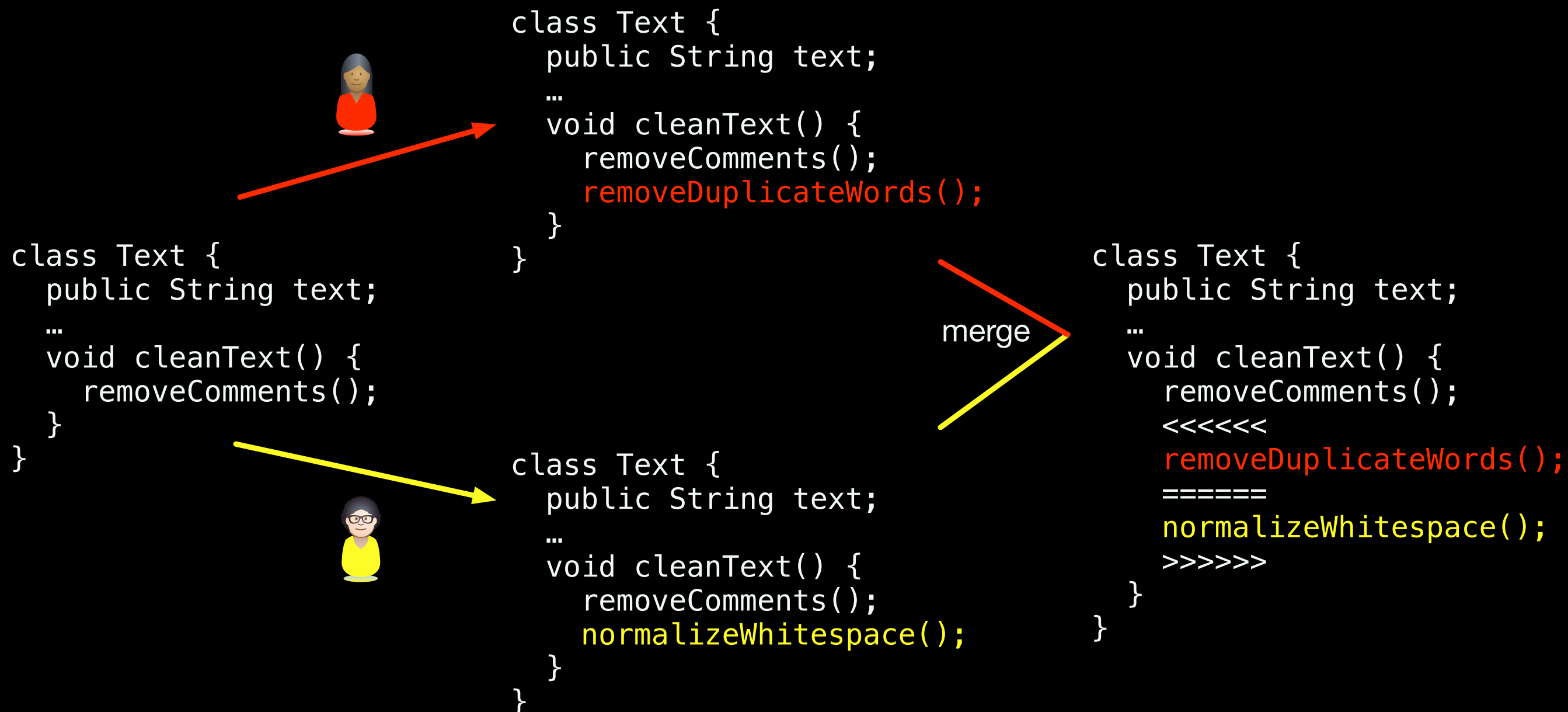
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Merge conflicts

(textual conflicts)

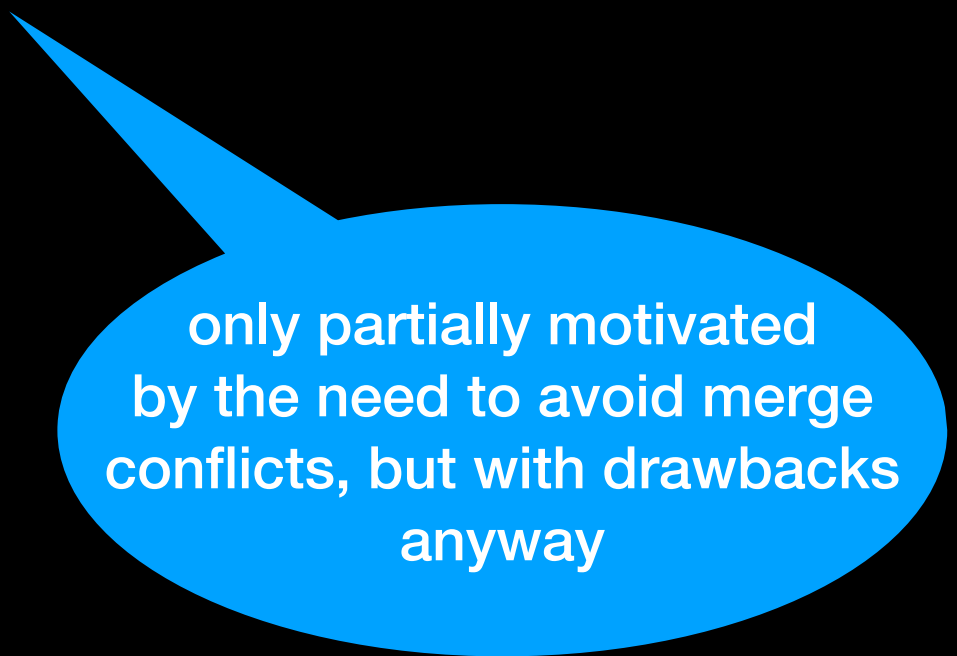


occur in **many** merge
scenarios [Kasi&Sarma, Brun et al,
Zimmermann]

affect **productivity** and
quality [Meyer et al]

Avoiding merge conflicts at any cost...

- ▶ rushing to finish changes first
- ▶ partial check-ins
- ▶ continuous integration
- ▶ trunk-based development
- ▶ feature toggles



only partially motivated
by the need to avoid merge
conflicts, but with drawbacks
anyway

Similar situation with
MVC

packages

module
structure
established
by the kinds
of classes

```
app
├── controllers
│   ├── articles_controller.rb
│   └── posts_controller.rb
```

```
models
├── article.rb
└── post.rb
```

```
views
├── article
└── posts
    ├── _table.html.erb
    ├── archived.html.erb
    ├── edit.html.erb
    ├── index.html.erb
    └── show.html.erb
```

post slice

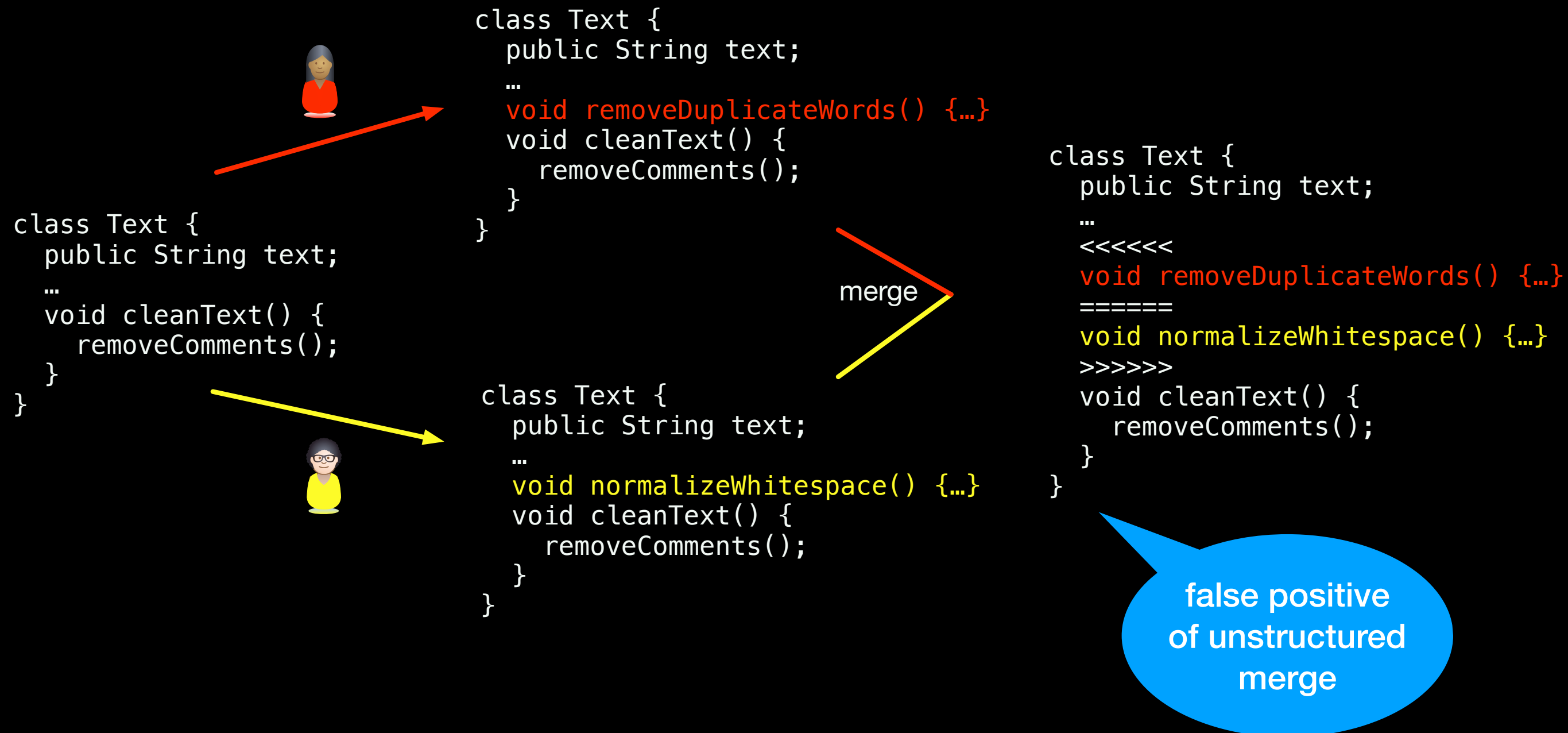
module
structure
established
by features
and tasks

Solution

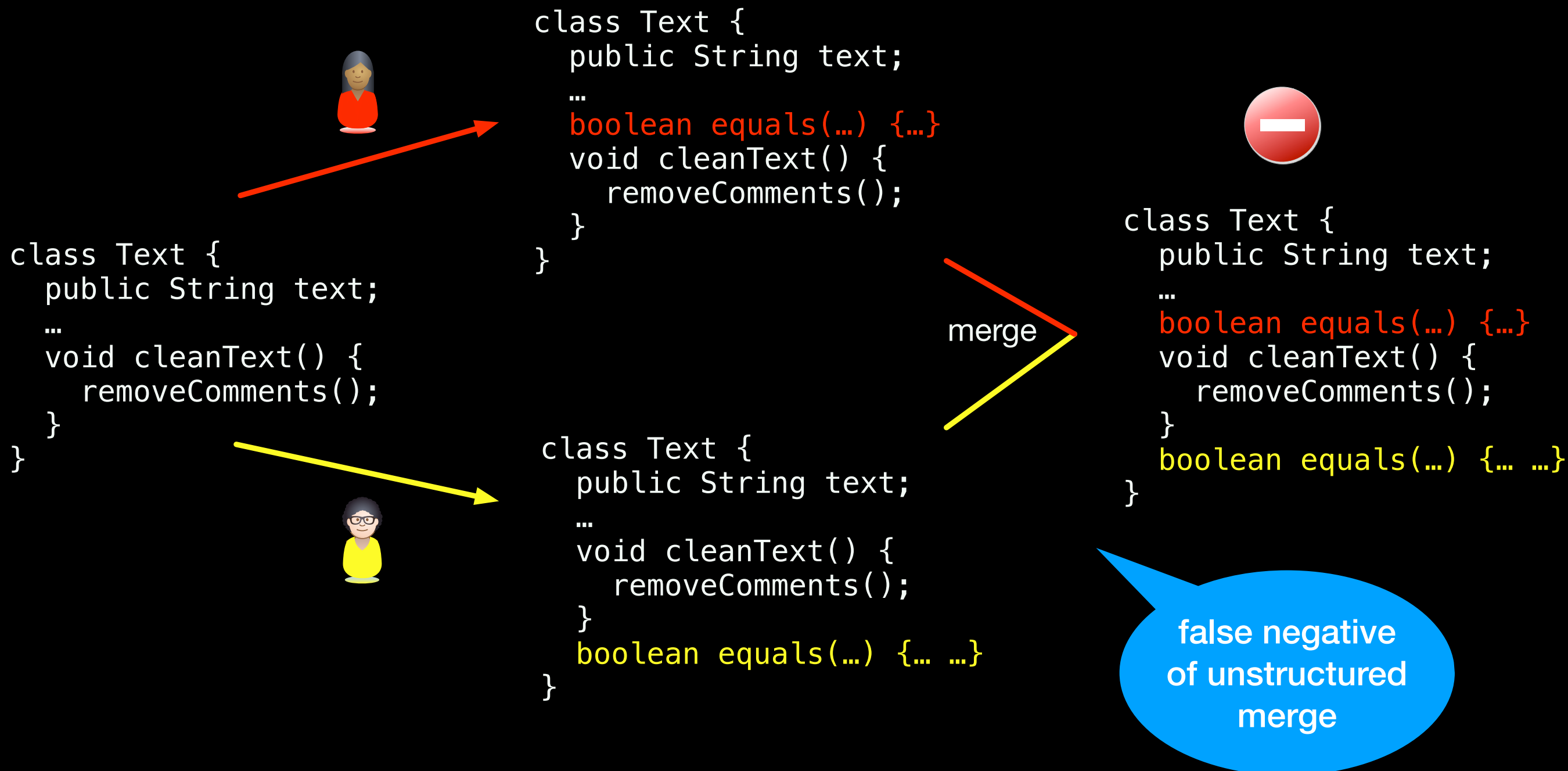
**How to reconcile
different modular
structures with
collaborative parallel
development?**

The code integration
process should be
**simpler and more
reliable**

Developers waste time by manually resolving conflicts that could be automatically solved



Current merge tools might integrate conflicting changes without warning developers



Substantial evidence about conflicts...

- Merge conflicts: 7-19% [Kasi&Sarma], 6-42% [Brun et al], 23-46% [Zimmermann]
- Build conflicts: 2-15% [Kasi&Sarma], 0.1-10% [Brun et al]
- Test conflicts: 6-35% [Kasi&Sarma], 3-28% [Brun et al]

Better tools for code
integration

Conflict avoidance,
detection and
resolution

Course details

Data	Aula, número	Aula, descrição	Formato
11/8	1	Apresentação do curso	Presencial
13/8	2	Gerência de configuração	Presencial
18/8	3	Merge textual	Presencial
20/8	4	Análise e transformação de programas, Merge estruturado	Presencial
25/8	5	Abordagens híbridas: merge semiestruturado, merge baseado em separadores	Presencial
27/8	6	Interferência e conflitos semânticos (estáticos e dinâmicos)	Presencial
1/9	7	Análise estática de código	Presencial
3/9	8	Detecção de conflito semântico com análise estática	Presencial
15/9	9	Geração de testes de unidade, entrega de roteiros	Presencial
17/9	10	Detecção de conflito semântico com testes	Presencial
22/9	11	Detecção de conflito semântico com análise dinâmica	Presencial
24/9	12	Merge semântico	Presencial
29/9	13	Uso de LLMs para merge e resolução de conflitos	Presencial
1/10	14	Revisão de código	Presencial
6/10	15	Uso de LLMs e agentes para revisão de código	Presencial
8/10	16	Definição do projeto	Presencial
13/10	17	Exame escrito, entrega de roteiros	Presencial
15/10	18	Projeto	Remota
20/10	19	Projeto	Remota
22/10	20	Projeto	Remota
27/10	21	Projeto	Remota
29/10	22	Projeto	Remota
3/11	23	Projeto, solução funcionando	Presencial
5/11	24	Projeto	Remota
10/11	25	Projeto	Remota
12/11	26	Projeto	Remota
17/11	27	Projeto	Remota
19/11	28	Apresentação do projeto	Presencial
24/11	29	Apresentação do projeto	Presencial
26/11	30	Apresentação do projeto	Presencial

Evaluation items

- Project (4.5)
- Exercise sets (1)
- Exam (4.5)
- Class participation (extra 0.5)

Mapping grade into evaluation level

> 9	A
> 7.5	B
> 6	C
< 6	D

Classroom code

nhvk7gj6

[**https://github.com/pauloborba/merge-and-code-review**](https://github.com/pauloborba/merge-and-code-review)



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